

Assignment 9: Time Series Analysis

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Spring 2026

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A09_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Check your working directory
 - Load the tidyverse, zoo, and trend packages
 - Set your ggplot theme

```
getwd()
```

```
## [1] "/home/guest/EDE_Spring2026"
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.5.1
## v ggplot2    3.5.1      v tibble     3.2.1
## v lubridate  1.9.3      v tidyr      1.3.1
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(zoo)
```

```
##  
## Attaching package: 'zoo'  
##  
## The following objects are masked from 'package:base':  
##  
##      as.Date, as.Date.numeric
```

```
library(trend)  
mytheme <- theme(  
  plot.title = element_text(size = 16, face = "bold"),  
  axis.title = element_text(size = 14)  
)  
theme_set(mytheme)
```

2. Import the ten datasets from the Ozone_TimeSeries folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named **GaringerOzone** of 3589 observation and 20 variables.

```
#1  
library(here)  
ozone10 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2010_raw.csv"), stringsAsFactors = FALSE)  
ozone11 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2011_raw.csv"), stringsAsFactors = FALSE)  
ozone12 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2012_raw.csv"), stringsAsFactors = FALSE)  
ozone13 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2013_raw.csv"), stringsAsFactors = FALSE)  
ozone14 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2014_raw.csv"), stringsAsFactors = FALSE)  
ozone15 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2015_raw.csv"), stringsAsFactors = FALSE)  
ozone16 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2016_raw.csv"), stringsAsFactors = FALSE)  
ozone17 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2017_raw.csv"), stringsAsFactors = FALSE)  
ozone18 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2018_raw.csv"), stringsAsFactors = FALSE)  
ozone19 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2019_raw.csv"), stringsAsFactors = FALSE)  
  
GaringerOzone <- rbind(ozone10, ozone11, ozone12, ozone13, ozone14,  
                      ozone15, ozone16, ozone17, ozone18, ozone19)  
dim(GaringerOzone)
```

```
## [1] 3589  20
```

Wrangle

3. Set your date column as a date class.
4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE.
5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame Days. Rename the column name in Days to "Date".

6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame `GaringerOzone`.

```
# 3
GaringerOzone <- GaringerOzone %>%
  mutate(Date = as.Date(Date, format = "%m/%d/%Y"))

# 4
GaringerOzone <- GaringerOzone %>%
  select(Date, Daily.Max.8.hour.Ozone.Concentration, DAILY_AQI_VALUE)

# 5
Days <- as.data.frame(seq(
  from = as.Date("2010-01-01"),
  to = as.Date("2019-12-31"),
  by = "day"
))
colnames(Days) <- "Date"

# 6
GaringerOzone <- left_join(Days, GaringerOzone, by = "Date")
summary(GaringerOzone)
```

```
##      Date      Daily.Max.8.hour.Ozone.Concentration DAILY_AQI_VALUE
## Min.   :2010-01-01   Min.   :0.00200                Min.    : 2.00
## 1st Qu.:2012-07-01   1st Qu.:0.03200                1st Qu. : 30.00
## Median :2014-12-31   Median :0.04100                Median  : 38.00
## Mean   :2014-12-31   Mean   :0.04163                Mean    : 41.57
## 3rd Qu.:2017-07-01   3rd Qu.:0.05100                3rd Qu. : 47.00
## Max.   :2019-12-31   Max.   :0.09300                Max.    :169.00
##      NA's      :63
```

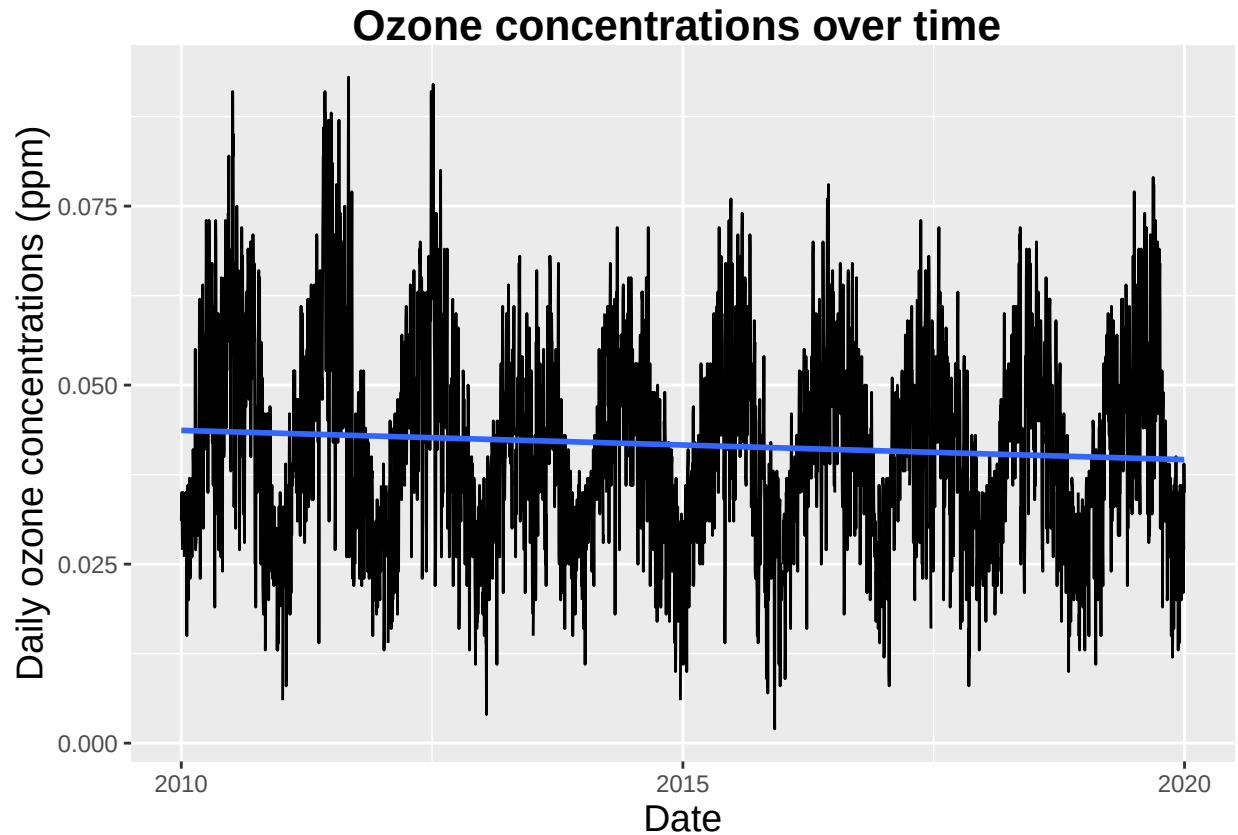
Visualize

7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

```
#7
ozone_line <- ggplot(GaringerOzone,
  aes(x=Date, y=Daily.Max.8.hour.Ozone.Concentration)) +
  geom_line() +
  geom_smooth(method = 'lm', se = FALSE) +
  labs(x = "Date",
    y = "Daily ozone concentrations (ppm)",
    title='Ozone concentrations over time ')
ozone_line
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 63 rows containing non-finite outside the scale range
## ('stat_smooth()').
```



Answer: Yes. The line plot shows a clear seasonal variation in ozone concentrations within a single year. The smoothed line shows a slight decrease in ozone concentrations over the period from 2010 to 2019. These indicate that the ozone concentrations remain fluctuated seasonally and have the trend of decreasing in general.

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

```
#8
GaringerOzone <-
  GaringerOzone %>%
  mutate(Ozone.clean = zoo::na.approx(Daily.Max.8.hour.Ozone.Concentration))
summary(GaringerOzone$Ozone.clean)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00200 0.03200 0.04100 0.04151 0.05100 0.09300
```

Answer:

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new `Date` column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

```
#9
GaringerOzone.monthly <- GaringerOzone %>%
  mutate(
    Year = year(Date),
    Month = month(Date)) %>%
  group_by(Year, Month) %>%
  summarise(
    Mean.Ozone = mean(Daily.Max.8.hour.Ozone.Concentration, na.rm = TRUE),
    .groups = "drop"
  )

GaringerOzone.monthly$Date <- ymd(
  paste(GaringerOzone.monthly$Year,
        GaringerOzone.monthly$Month,
        "01",
        sep = "-")
)
```

10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

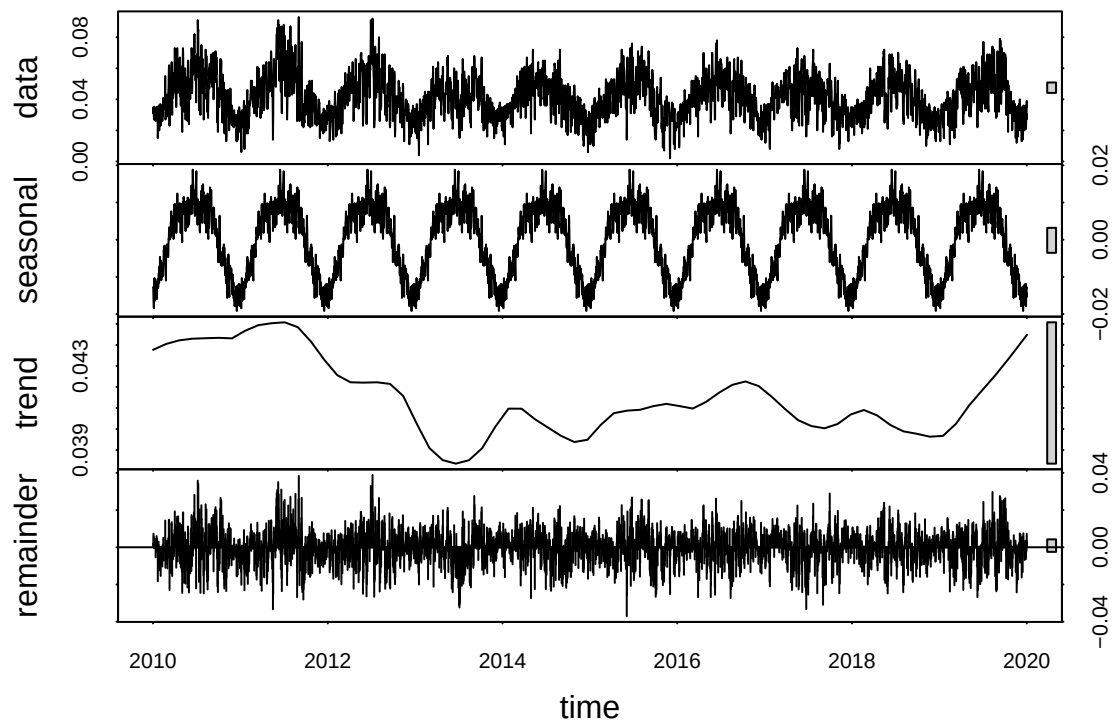
```
#10
f_month = month(first(GaringerOzone$Date))
f_year = year(first(GaringerOzone$Date))

GaringerOzone.daily.ts <- ts(
  GaringerOzone$Ozone.clean,
  start = c(f_year, f_month),
  frequency = 365
)

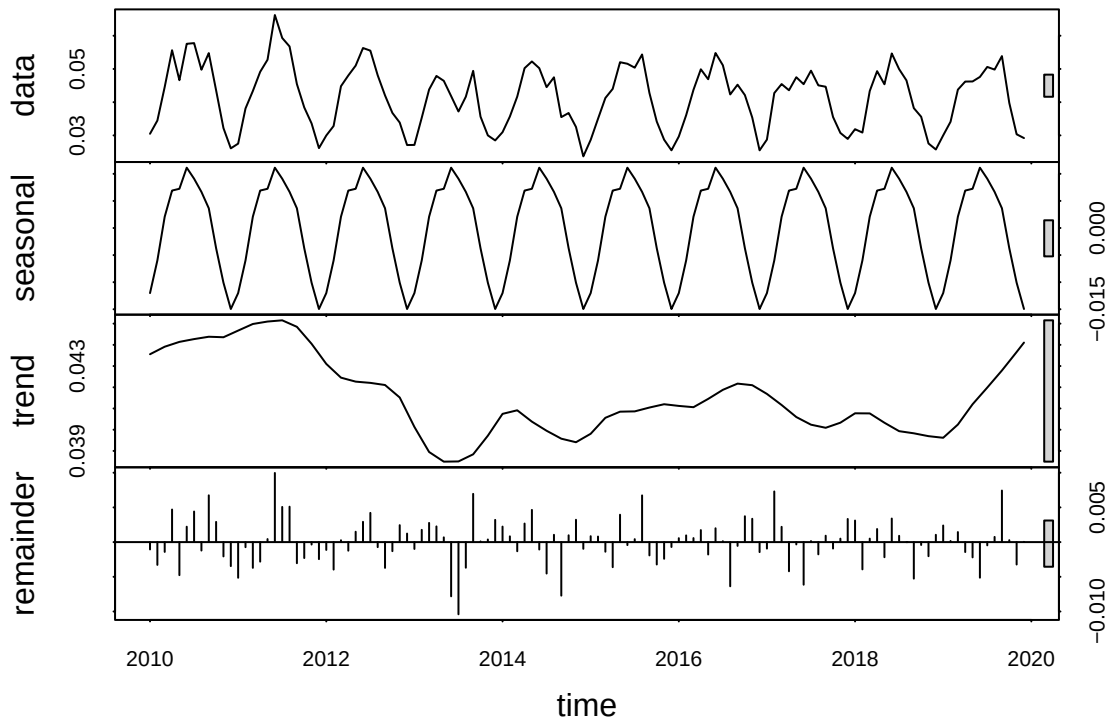
GaringerOzone.monthly.ts <- ts(
  GaringerOzone.monthly$Mean.Ozone,
  start = c(f_year, f_month),
  frequency = 12
)
```

11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

```
#11
GaringerOzone.daily.decomp <- stl(GaringerOzone.daily.ts, s.window = "periodic")
GaringerOzone.monthly.decomp <- stl(GaringerOzone.monthly.ts, s.window = "periodic")
plot(GaringerOzone.daily.decomp)
```



```
plot(Garinger0zone.monthly.decomp)
```



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

```
#12
library(Kendall)

GaringerOzone.SMK <- SeasonalMannKendall(GaringerOzone.monthly.ts)
GaringerOzone.SMK
```

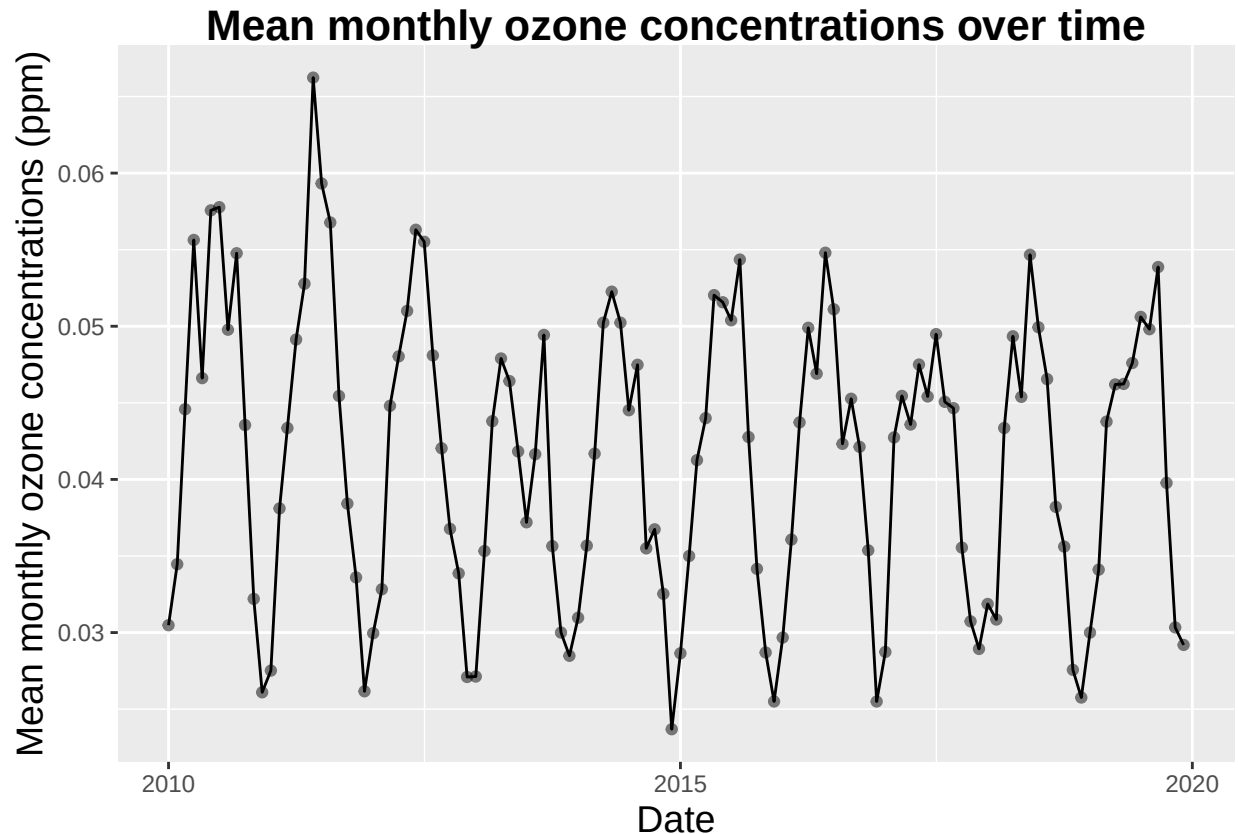
```
## tau = -0.163, 2-sided pvalue =0.022986
```

Answer: The seasonal Mann-Kendall test is most appropriate because the monthly ozone series shows strong seasonality and varies substantially by month. This test evaluates if there is a monotonic long-term trend and also considering seasonal patterns, which a standard Mann-Kendall test may ignore.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a `geom_point` and a `geom_line` layer. Edit your axis labels accordingly.

```
# 13
mean_monthly_ozone <- ggplot(GaringerOzone.monthly,
                             aes(x=Date, y=Mean.Ozone)) +
  geom_point(alpha = 0.5) +
  geom_line() +
```

```
labs(x = "Date",
     y = "Mean monthly ozone concentrations (ppm)",
     title = "Mean monthly ozone concentrations over time")
mean_monthly_ozone
```



14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

Answer: Mean monthly ozone concentrations show a slight but statistically significant decreasing trend over the 2010s, even with a strong seasonal variability. The Seasonal Mann-Kendall test confirms a negative monotonic trend after accounting for seasonality ($\tau = -0.163$ indicates a low to moderate decreasing trend, and 2-sided p-value = $0.023 < 0.05$, which is statistically significant).

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the `EnoDischarge` on the lesson Rmd file.
16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

```
#15
GaringerOzone.monthly.noseason <- GaringerOzone.monthly.ts -
  GaringerOzone.monthly.decomp$time.series[, "seasonal"]
```


#16

```
GaringerOzone.MK.noseason <- MannKendall(GaringerOzone.monthly.noseason)
GaringerOzone.MK.noseason
```

```
## tau = -0.179, 2-sided pvalue =0.0037728
```

Answer: After using Mann-Kendall test that removes the effect of seasonality, the mean monthly ozone concentration shows a stronger and more statistically significant negative trend ($\tau = -0.179$, $p = 0.0038$) compared to the Seasonal Mann-Kendall test ($\tau = -0.163$, $p = 0.023$). This suggests that seasonal variability was partially moderating the long-term decline in ozone concentrations. Once the seasonal component is removed, the decreasing trend becomes more prominent.